

AMD Instinct MI400 Series GPU Architecture

Powering the next frontier of AI
and High Performance Computing

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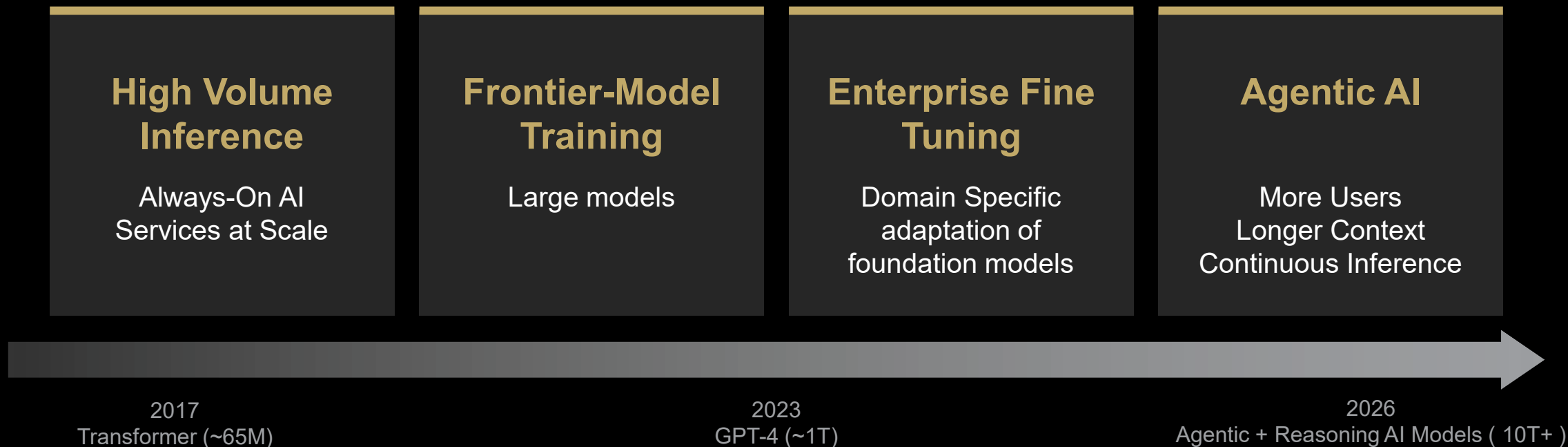
Senior Fellow, Graphics Architecture, AMD

Hot Chips

The AMD Instinct logo is displayed on a white rectangular background. It features the word "AMD" in a bold, sans-serif font, followed by a stylized square icon containing a triangle. Below this, the word "INSTINCT" is written in a smaller, all-caps, sans-serif font.

AMD
INSTINCT

AI Is Driving The Next Era Of Computing



Infrastructure Needs to Move Data as Fast as Models Scale

AMD Helios™

The Rackscale AI Infrastructure

2.9 Exaflops
AI Compute

31 TB
HBM4 Memory

1.7 PB/s
HBM4 Bandwidth

72 GPUs
Per Rack

260 TB/s
Scale Up Bandwidth

43 TB/s
Scale Out Bandwidth



The Building Block of AMD Helios™

Compute Tray – Combining compute, host, memory and networking

GPU Compute & Scale Up

4 AMD Instinct™ MI455X EAMs
36 UALoE Links (x2) per EAM

CPU Host

Single Socket AMD EPYC™ 9006 SP7 Server CPU
Coherent AMD Infinity Fabric™ (x16) to MI455X

Scale Out Network

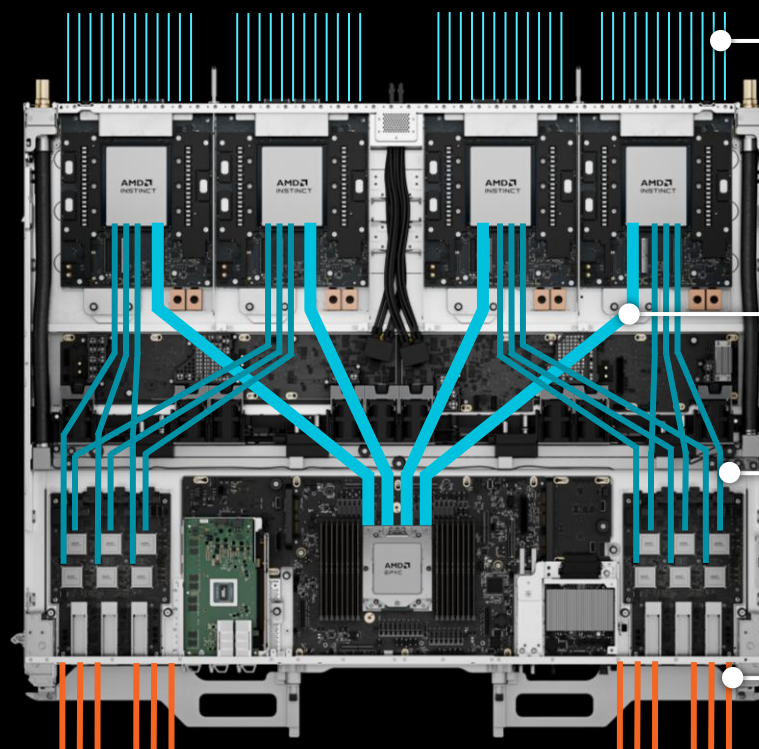
Up to 3x AMD Pensando™ Vulcano 800 AI NIC Per EAM
UALink128 (x8) to MI455X EAM

Front-End Network

AMD Pensando™ Salina 400 DPU FH PCIe CEM NIC

3 UALoE Links (x2) UALink128 Link (x8) Infinity Fabric Link (x16)

Scale Up UALoE Network



UALoE to scale-up:
1.8 TB/s/dir per GPU

Infinity Fabric to CPU:
128 GB/s/dir per GPU

UALink to NICs:
128GB/s/dir per NIC
Vulcano 800 AI NIC

800GbE
Ethernet

Scale Out Ethernet Network

AMD Instinct™ MI455X GPU

Enhanced Modular Chiplet Architecture

2x Fabric and Cache Die (N3P)

192 Channel HBM4 Interface

192 MB Global L2 Cache

2x I/O Die (N3P)

2x PCIe Gen 6 or 3x AMD AI-NICs using UALink

Infinity Fabric, 256 GB/s BW (bi-dir)

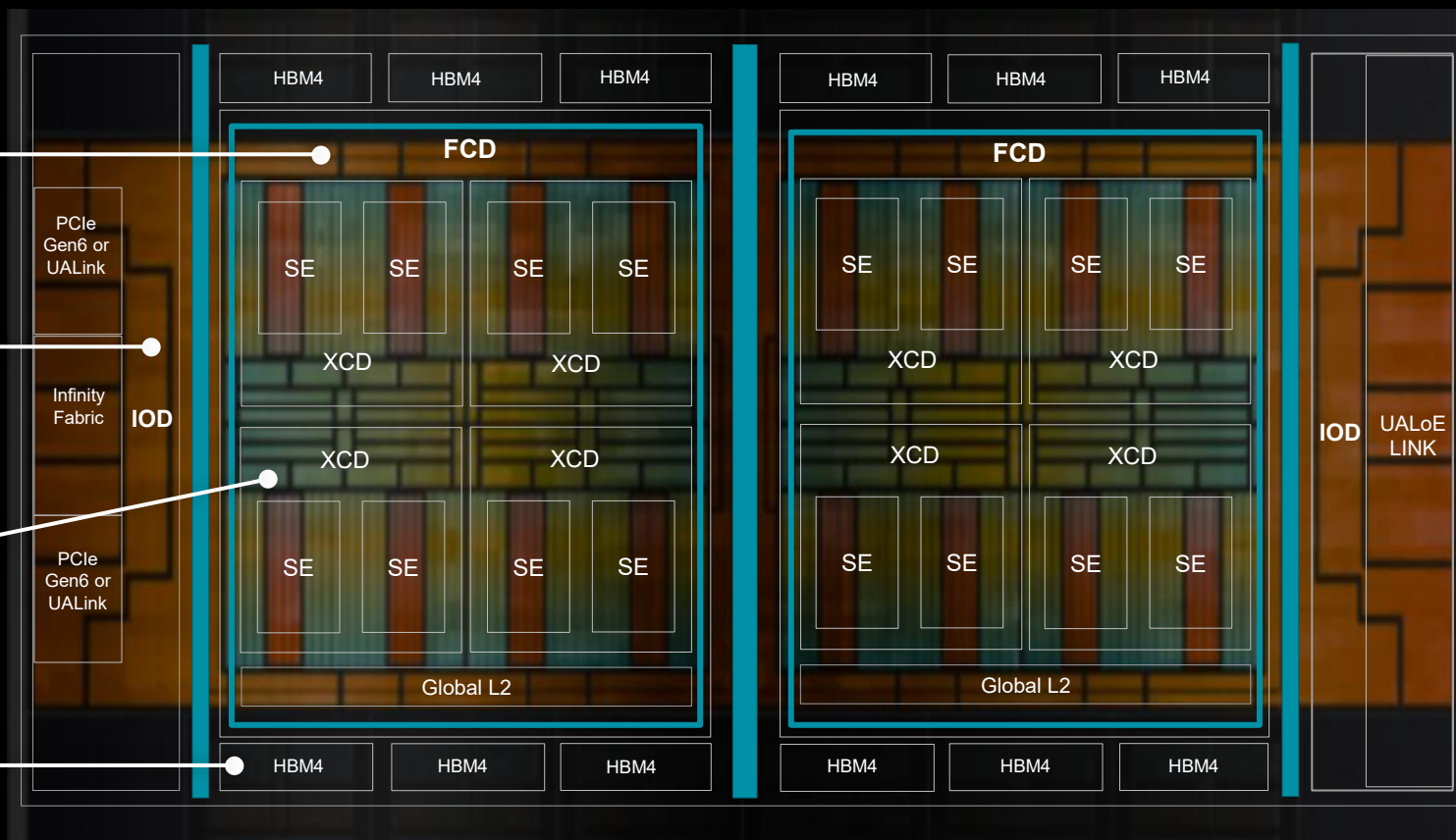
72 UALoE lanes, 3.6 TB/s BW (bi-dir)

8x Accelerator Complex Die (N2)

256 Total Active Work Group Processors

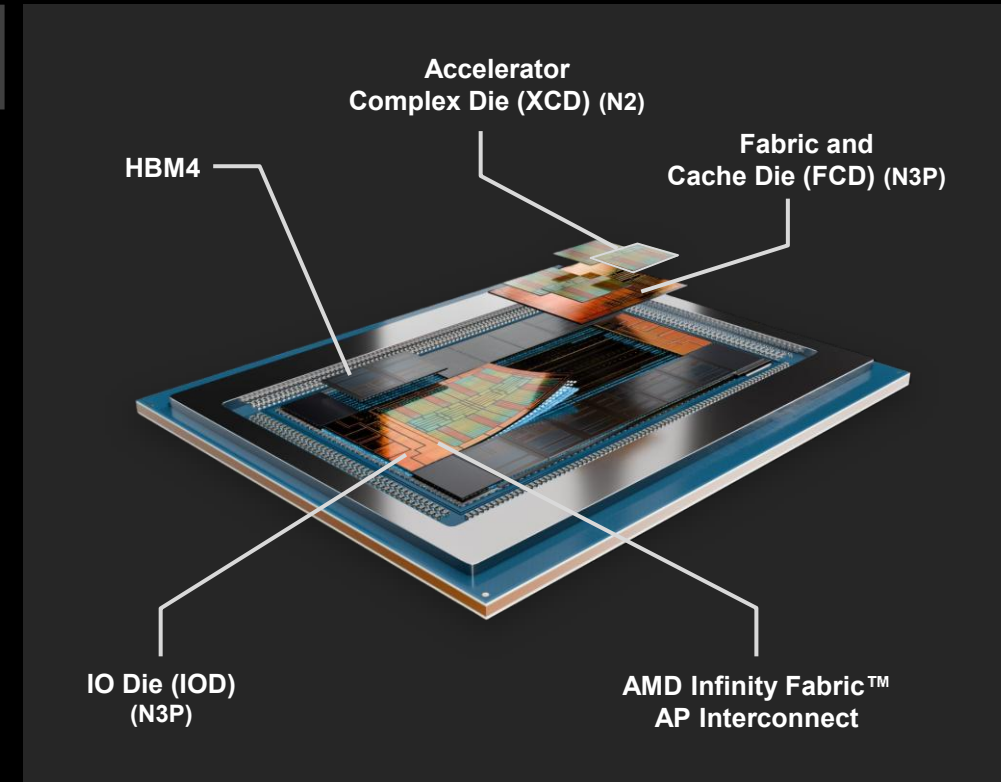
12x HBM4 Memory

432 GB at 23.3TB/s



Extended Packaging Leadership

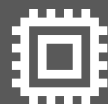
Innovation	What's New	AI Workload Impact
Chiplet Architecture	Compute, memory, cache, and I/O functions are partitioned across specialized dies	Optimizes each function independently for performance, power, and manufacturability
3D Hybrid Bonded XCDs	Compute dies stacked and connected through high-density die-to-die interconnects	Higher compute density and improved performance-per-watt
HBM4 Integration	12 HBM4 stacks 432 GB capacity 23.3 TB/s bandwidth	Larger models, KV caches, and activations remain resident in local memory
CoWoS-L Advanced Packaging	High-bandwidth on-package communication between compute, memory, and I/O dies	Low latency and high bandwidth enable both GPU-wide single operators and sharding techniques like KV cache parallelism, TP, and EP



Advanced Packaging Enables the Next Generation of AI Scaling

Key Innovations

Upgraded Memory



Enhanced Cache/Memory hierarchy for growing models, context windows, and KV caches

- 432 GB HBM4
- 23.3 TB/s bandwidth
- 192 MB L2 – Larger cache hierarchy
- Expanded local storage

LARGER MODELS AND
LONGER CONTEXT STAY LOCAL

Enhanced Compute



Improved compute for exponential demand of AI workloads

- Higher IPC and Fine-grained scheduling
- MX formats with fractional scaling
- Transcendental Engine
- Increased vector throughput WGP

IMPROVED TRAINING & INFERENCE
PERFORMANCE PER WATT

Improved Efficiency



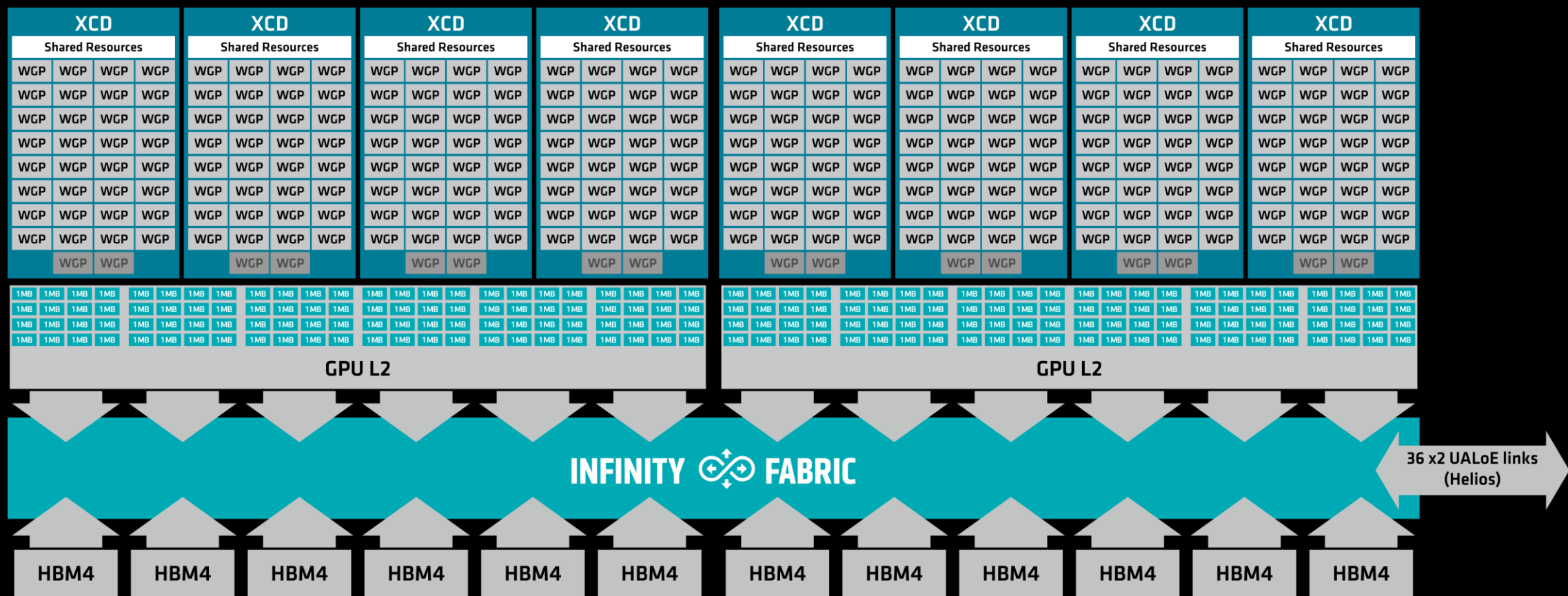
Reduced data movement, and synchronization increase delivered performance

- Tensor Data Mover
- Topology Aware DMA
- L2 Multicast
- Lower dispatch latency

IMPROVED UTILIZATION AND
REDUCED MEMORY OVERHEAD

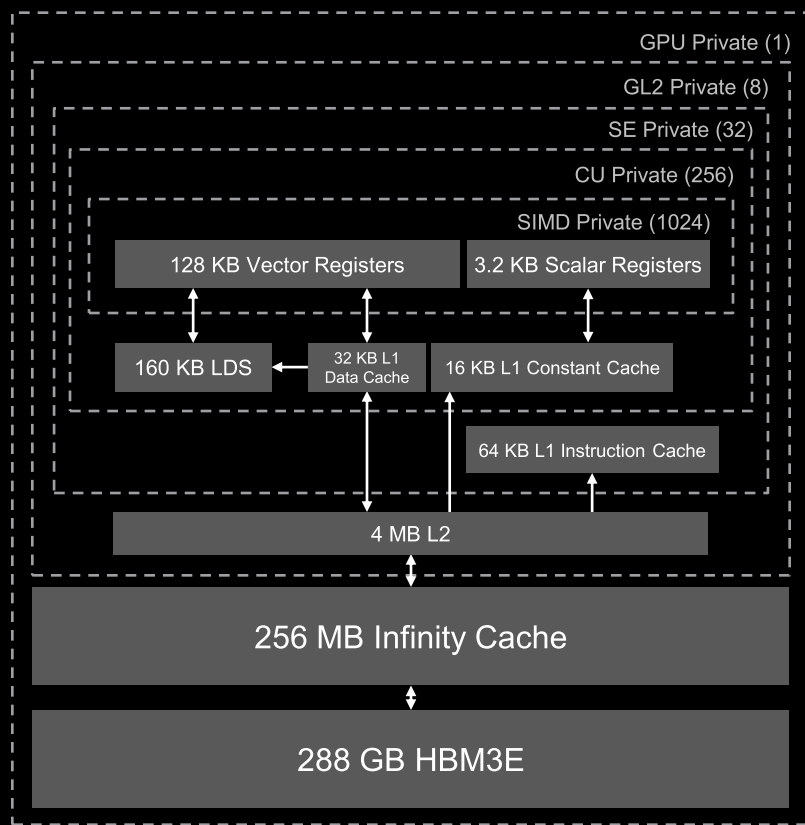
AMD Instinct™ MI455X GPU

Conceptual SoC Organizational Block Diagram

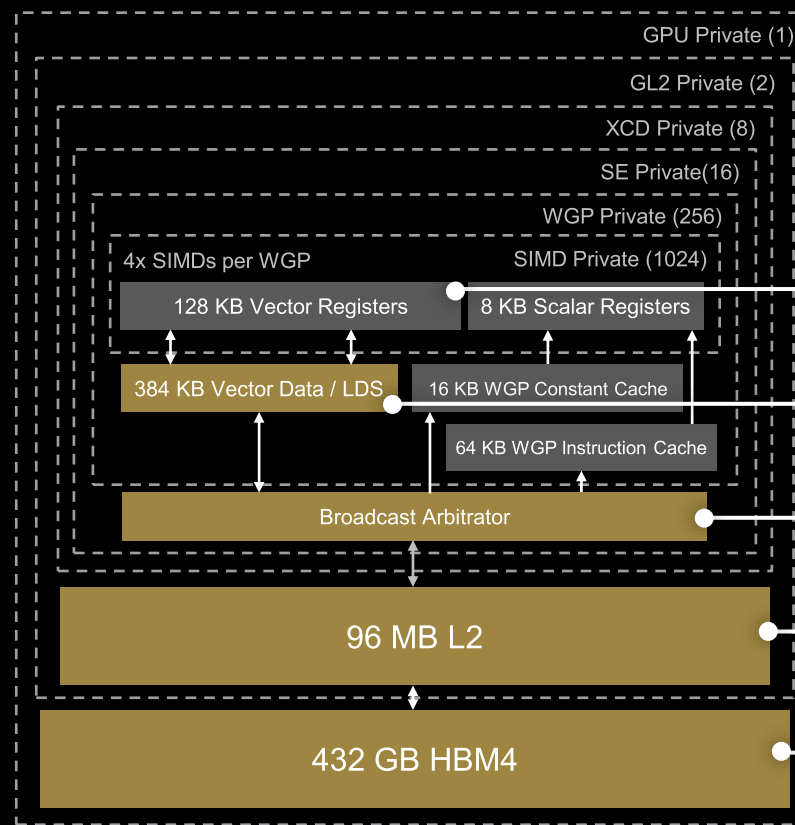


Upgraded Cache & Memory

AMD Instinct™ MI355X



AMD Instinct™ MI455X



Bandwidth Improvements

2.0x VGPR per SIMD

2.0x LDS per WGP / CU

up to 4x BW amplification

1.5x per L2 vs  Cache

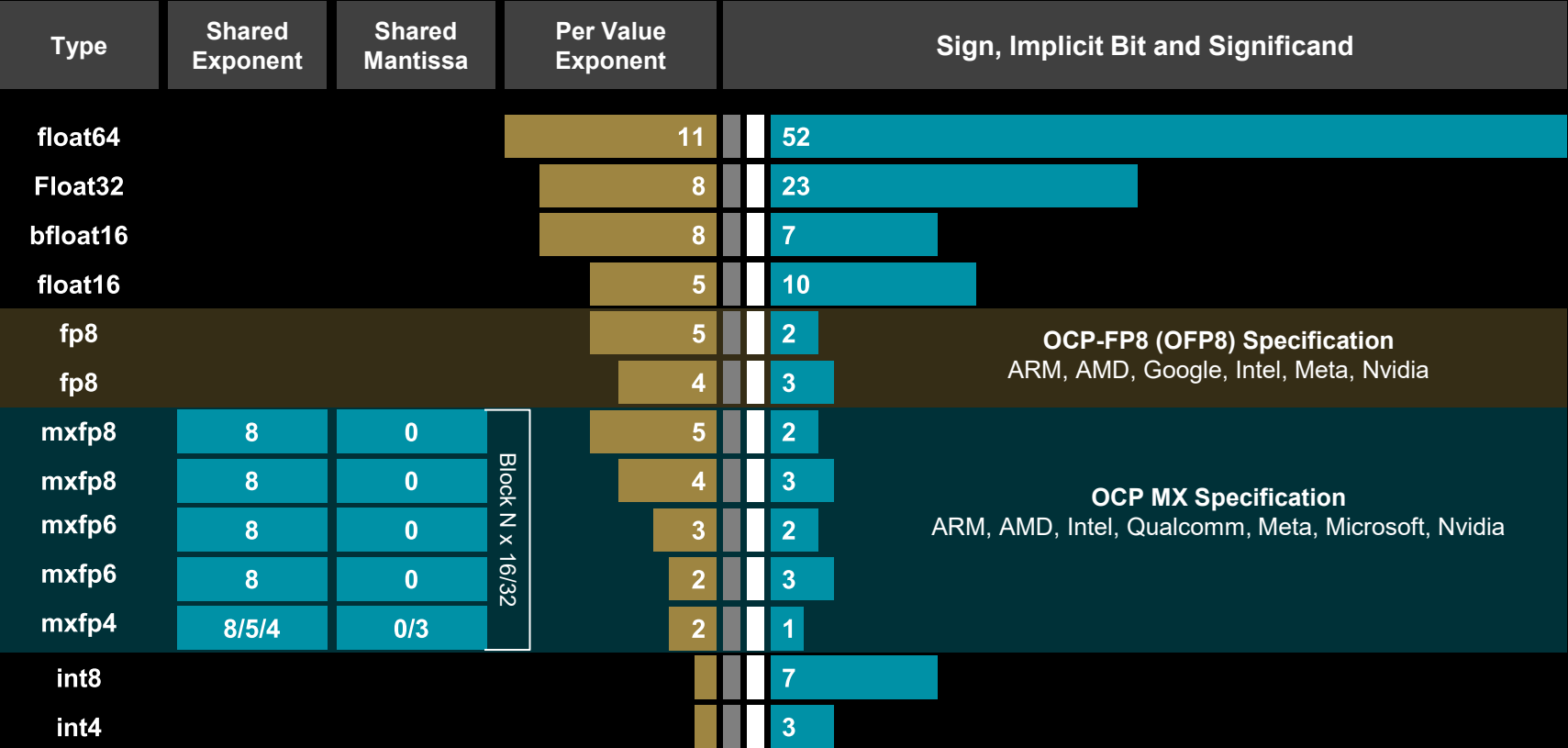
2.9x total HBM

Enhanced Compute

Innovation	What's New	AI Workload Impact	Peak Computation	Instinct™ MI455X GPU	vs Instinct™ MI355X GPU
Higher IPC	Native Wave32 execution, single cycle execution on 32 wide vector compute machine	Reduces instruction latency, branch divergence penalties, register pressure. Better suited for latency sensitive compute.	OCP MXFP4	40.26 PF	Up to 4X
Advanced AI Datatypes	Adds MXFP8, MXFP6, MXFP4 with block-scale 16/32 and fractional scaling for MXFP4	Block-level fractional scaling enables practical FP4 training, reducing quantization error vs. full-tile scaling	OCP MXFP6	20.13 PF	Up to 2X
Enhanced Transcendental Engine	New tanh instruction and improved transcendental throughput	Faster Softmax, activations, and attention pipelines	OCP MXFP8/FP8	20.13 PF	Up to 4X
Larger Compute Resources	256 WGP architecture with higher per-WGP throughput	More compute delivered per clock cycle	Matrix FP16/BF16	5.03 PF	Up to 2X
			Vector FP16	315 TF	Up to 2X
			Matrix/Vector FP32	315 TF	Up to 2X

AMD Instinct™ MI455X Optimized Low-Precision Support for AI Execution

Optimized Compute For AI



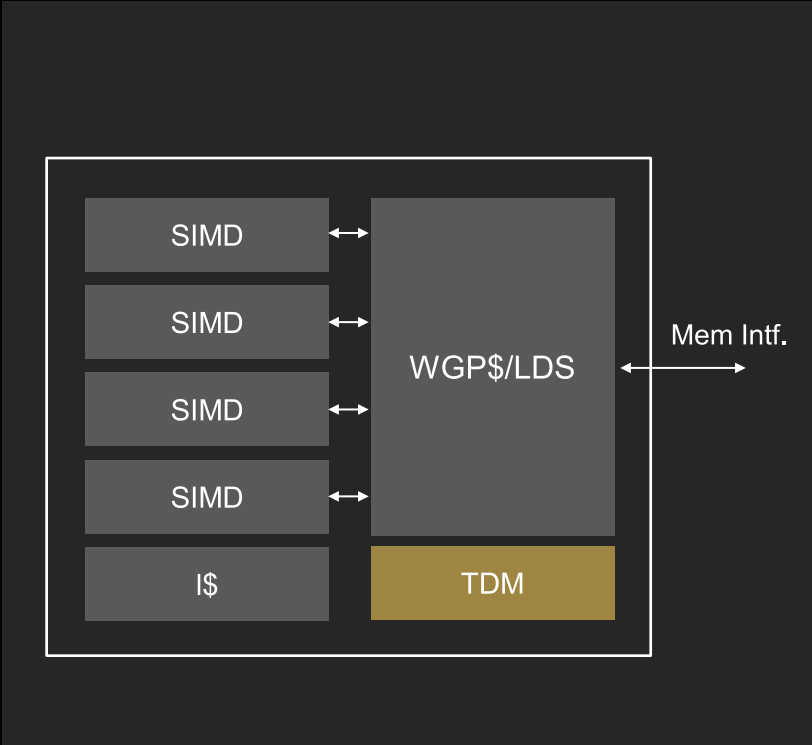
Blocks sharing a common scale factor may now be 16 or 32 elements in size, and a new fractional scale is available for MXFP4 data.

4-bit tensor LUT instructions allow 4-bit memory storage and conversion to 4/6/8-bit MXFP compute formats.

Improved Efficiency

Less Data Movement and Higher Utilization

Innovation	What's New	AI Workload Impact
Tensor Data Mover (TDM)	Direct async Global ↔ LDS transfers without register staging	Overlaps memory movement and compute
Work group Cluster, Multi-cast, Data Prefetch	Multiple WGP's form a cluster (Work group Cluster), L2 data return multicast, prefetch data to L2 to hide memory latency	Together they reduce redundant traffic and drive the efficiency of operators like GEMM, Flash Attention, while enabling the wave-specialized kernels that are becoming dominant
Split / Named Barriers	More efficient synchronization	Less waiting between work phases
Lower Dispatch Latency	New command processor and scheduling	Reduces dispatch overhead and idle gaps so short kernels stay fed and overlap cleanly



Reduce Memory Traffic and Idle Cycles to Increase Delivered Performance

Hardware Accelerated Data Movement

Dedicated DMA Engines Per GPU

Hardware engines move data independently of WGPs
Data transfers execute in parallel with AI kernels
Supports local & remote GPU memory transfers

Key Benefits

Topology-Aware DMA Engines

DMA engine traffic automatically affinitizes traffic to UALoE links, allowing software to be oblivious to data placement

Overlap Compute and Communication

DMA moves data while WGPs continue executing AI kernels

Efficient Remote Memory Access

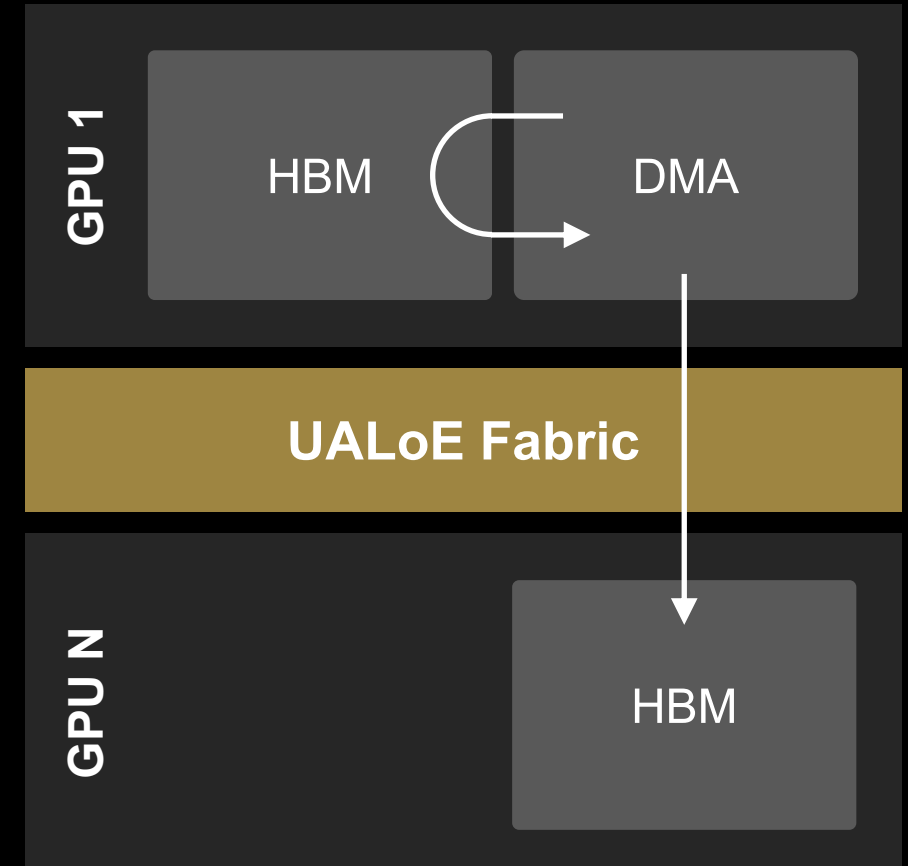
Direct Data transfers between GPUs across the UALoE fabric

Scales Across 72 GPUs

DMA automatically distributes traffic across available fabric resources

Reduces Fabric Congestion

Front-end / back-end DMA architecture distributes traffic intelligently





AI-Driven Development Platform

Accelerating development velocity and performance optimization

AI SKILLS Popular agents become ROCm superusers	Skills	 Claude  Codex  Cursor  Gemini
AI ASSISTED OPTIMIZATION AI that optimizes workloads	Hyperloom	End-to-end AI Workload Optimization
ROCm CORE Compilers, tools, libraries, frameworks	Framework Integration	    
	Core SDK	Libraries
		Compilers and Tools
		Runtime

ROCm.AI Delivers Leadership Performance on MI455X

MI455X VS MI355X

MEMORY	COMPUTE	SCALE UP NETWORKING	SCALE OUT NETWORKING
20 TB/s Measured MLA (Decode, FP8)	20 PF Measured FP4 Compute	3.2 TB/s Measured Bandwidth	190 GB/s Measured Bandwidth
3.8x More Performance	3.3x More AI Compute	3.5x More Scale Up Bandwidth	2x More Scale Out Bandwidth

AMD reached an estimated **2.4x** increase in AI energy efficiency with **MI455X** over MI355X, building momentum toward its **2030 goal to deliver a 20x increase** in rack-scale efficiency for AI training and inference

The Future of AI Runs on AMD

1 

Advanced Packaging Leadership

Modular architecture for heterogeneous integration of SOIC, HBM4, & COWOS-L

2 

Optimized AI Throughput & Utilization

Maximizing AI performance through efficient compute and data movement

3 

Hardware Confidential Computing

Model Confidentiality and Integrity preserved at GPU & rack scale

4 

Upgraded Memory Subsystem

Expanded cache hierarchy and HBM4 bandwidth enable better tokens/s/\$

5 

72 GPU Rack-scale Architecture

Enables trillion parameter models and rack-scale AI deployment

6 

Fault Tolerant Scale Up Design

Designed for continuous operation at rack scale

Built for Frontier Training and High-Volume Inference at Rack Scale



ENDNOTES

MI400-006: "Based on AMD Performance Labs calculations (June 2026) using an AMD Instinct™ MI455X GPU, peak theoretical precision performance (FP32, FP16, BF16, MXFP6, MXFP8, FP8, MXFP4 Matrix/Vector), compared to published specifications for AMD Instinct™ MI355X, MI350X, MI325X, MI300X, MI250X, and MI100 GPUs. Results may vary by system configuration and datatype. MI400-006"

MI400-008: "Calculations by AMD Performance Labs in June 2026, based on the published memory capacity and memory bandwidth specifications of an AMD Instinct™ MI455X GPU vs the published memory capacity and memory bandwidth specifications of AMD Instinct™ MI355X, MI350X, MI325X, MI300X, MI250X and MI100 GPUs, respectively. System manufacturers may vary configurations, yielding different results. MI400-008"

MI400-027 Based on testing and calculations by AMD Performance Labs in July 2026, on systems configured with a single AMD Instinct™ MI455X GPU powered by AMD ROCm.ai vs an AMD Instinct™ MI355X GPU to determine median bandwidth in TB/s using an industry standard Multi-Latent Attention (MLA) benchmark. System manufacturers may vary configurations, yielding different results. MI400-027

MI400-028: Based on calculations by AMD Performance Labs in July 2026, on a system configured with a single AMD Instinct™ MI455X GPU powered by AMD ROCm.ai to determine MXFP4 performance using an AITER GEMM kernel benchmark compared to the published specifications for a single AMD Instinct™ MI355X GPU. System manufacturers may vary configurations, yielding different results.

MI400-029: Based on calculations by AMD Performance Labs in July 2026, on systems configured with 4x AMD Instinct™ MI455X GPUs and 8x AMD Instinct™ MI355X GPUs, both powered by AMD ROCm.ai, to determine single GPU scale-up networking bandwidth using a common memory bandwidth benchmark with a 16GB transfer size. Median results from five runs are reported. System manufacturers may vary configurations, yielding different results.

MI400-030: Based on testing by AMD Performance Labs in July 2026, on systems powered by AMD ROCm.ai and configured with an AMD Instinct™ MI455X GPU and AMD Pensando™ Vulcano 800 AI NIC and AMD Instinct™ MI355X GPU with AMD Pensando™ Pollara 400 AI NIC to determine bidirectional scale-out networking bandwidth running a bandwidth benchmark. System manufacturers may vary configurations, yielding different results.